Fusion power for quantum computing: build the load model first

Executive investment and research pitch | Beyond Q global comparison

Version 2.3-PITCH | 29 October 2026 | Independent proposed research; not endorsed by MIT, Microsoft, Google, CFS or any Chinese institution.

Evidence code: SOURCE-TRACED = sourced statement; VERIFIED-DERIVED = checked arithmetic; POTENTIAL-DERIVED = hypothetical-input result; UNRESOLVED = not sufficiently evidenced; LAB-MEASURED reserved for a traceable physical measurement and scope.

Decision in one paragraph

Recommend a staged, independent feasibility study—not immediate construction—of quantum-first infrastructure powered by commercial D-T fusion and, only if justified, a D-D isotope-production companion. Assess Google superconducting and neutral-atom systems, Microsoft topological and partner-neutral-atom routes, and Chinese superconducting, neutral-atom, and photonic architectures. Compare colocated campus facilities against embedded quantum nodes in existing classical data centers. The initial objective is to discover whether isotope scarcity is a binding constraint and whether an ARC-derived D-D unit can produce qualified He-3/T at a competitive all-in cost.

Why now

- Real customer anchor, not a quantum contract: Google signed for 200 MW from the first planned CFS ARC plant; the 400 MW plant output is a design target, with grid delivery projected in the early 2030s. [R03,R23]
- Quantum architectures are diverging: Willow demonstrated below-threshold surface-code behavior; Google added neutral atoms in 2026. Microsoft reported substantial Majorana 2 improvement and supports partner neutral-atom logical systems. [R08,R09,R12,R14]
- China supplies a valuable external scale comparator: Hanyuan-2 was reported below 7 kW for a 200-atom neutral-atom machine, while Origin SL3000 specifies 3 mW cold capacity at 100 mK but does not disclose whole-wall-plug power. [R15-R18]
- A 2026 peer-reviewed ARC V3A paper targets about 1.13 GW fusion power and 400 MW net electricity. It separately counts about 100 MW blanket exothermic heat and 20 MW injected heating in that D-T design. These numbers are not portable to hypothetical D-D operation. [R01,R02]

Architecture choices to compare

Case	Configuration	Decision question
C1 power-max	2 × commercial D-T ARC reference	What is the value of electricity and ordinary D-T fuel-cycle operation?
C2 Beyond Q	1 × D-T + speculative D-D isotope unit	Can incremental isotope/resilience value justify foregone electricity?
C3 grid-hybrid	1 × D-T + D-D + grid purchases/exports	Could flexible purchasing outperform dedicated internal supply?
C4 embedded node	Quantum hardware in existing classical data center	Is separate fusion construction unnecessary at current quantum

		hardware scale?

Preliminary screening result — not a reactor prediction

With an illustrative 20 MW primary D-D power, 28% in-plasma tritium secondary burn, 50% effective Li-6 capture, 80 MW delivered RF (70% wall-plug), 15 MWe other auxiliaries, 90% RF deposition, 85% thermal collection and 40% conversion, our independent algebraic model gives 13.5 MW secondary D-T, 8.4 MW Li-6 Q energy, 38.7 MWe gross generation, and -90.6 MWe net for the D-D unit. Every operating-point and efficiency assumption is unvalidated for ARC-D-D. No neutron-transport or plasma solver has been run.

The conditional difference versus a second 400 MWe D-T unit is 490.6 MW of electrical capability. At an illustrative $65/MWh and 85% capacity factor, this corresponds to approximately $237.4 million/year in foregone energy value before CAPEX, commodity offtake, and market effects. This is an opportunity-cost benchmark, not a cost forecast. The D-D unit is being tested as an isotope refinery, not presumed to be a net power generator.

Recommended ask and stage gates

1. Commission an independent QPU-to-wall-plug demand model for the three technology ecosystems, with measured cooling and decoder power requested directly from operators.
2. Commission a plasma-physics feasibility review of an ARC-class D-D operating point and a neutron-transport design study for the proposed FLiBe blanket.
3. Publish validated isotope, heat, electricity and economic ledgers with failing cases retained; conduct third-party review before proposing hardware or requesting partnership.

What would change the answer?

A valid D-D operating point with demonstrated isotope recovery, a measured He-3 demand bottleneck, a competitive lifecycle cost versus existing supply, and a positive comparison against a second conventional D-T plant would justify proceeding. If any gate fails, stop the D-D build proposition while retaining the quantum demand and heterogeneous-node research.

Three questions for an MIT-affiliated technical reviewer

- Which experimentally anchored D-D confinement and current-drive regime could fit ARC V3A magnet/vessel/divertor constraints, if any?
- Which neutron-transport and fuel-cycle calculations are necessary to validate tritium balance and blanket heat before techno-economics?
- Which quantum hardware cooling and reliability metrics would be necessary to make He-3 supply, rather than chip manufacture or control, the limiting scale factor?

No partnership, academic endorsement, or verified retrofit feasibility is claimed. Full citations, algebraic code, assumptions, and acceptance gates follow in the companion technical paper. [R01-R27]